



LOG ANALYSIS PRACTICE

Lab Documentation Report

Tools: Windows Event Viewer | LECmd | LogParser Lizard | Elastic
SIEM

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1. Lab Overview

This lab focused on analyzing Windows security logs and browser history artifacts to identify suspicious activity. Failed login attempts were investigated using Windows Event Viewer, while browser history was analyzed using Eric Zimmerman's LECmd tool. The objective was to detect brute-force login activity and identify potentially malicious URLs visited on the system.

Field	Details
Lab Activity	Log Analysis Practice
Windows Host	Windows VM
Attacker IP	192.168.56.101
Monitoring Platform	Elastic SIEM
Event ID Investigated	4625, 7045
Browser Analysis Tool	LECmd

2. Windows Event Log Analysis

Event ID 4625 – Failed Login Attempts

Event ID 4625 indicates a failed login attempt on a Windows system. Multiple failed login attempts occurring within a short period may indicate brute-force activity.

Investigation Procedure

1. Open Event Viewer.
2. Navigate to Windows Logs → Security.
3. Apply filter for Event ID 4625.
4. Review timestamps and usernames.
5. Identify repeated login failures.

Findings

Multiple failed login attempts were detected against the Windows VM. The repeated authentication failures suggested a potential brute-force attack simulation originating from the testing environment.

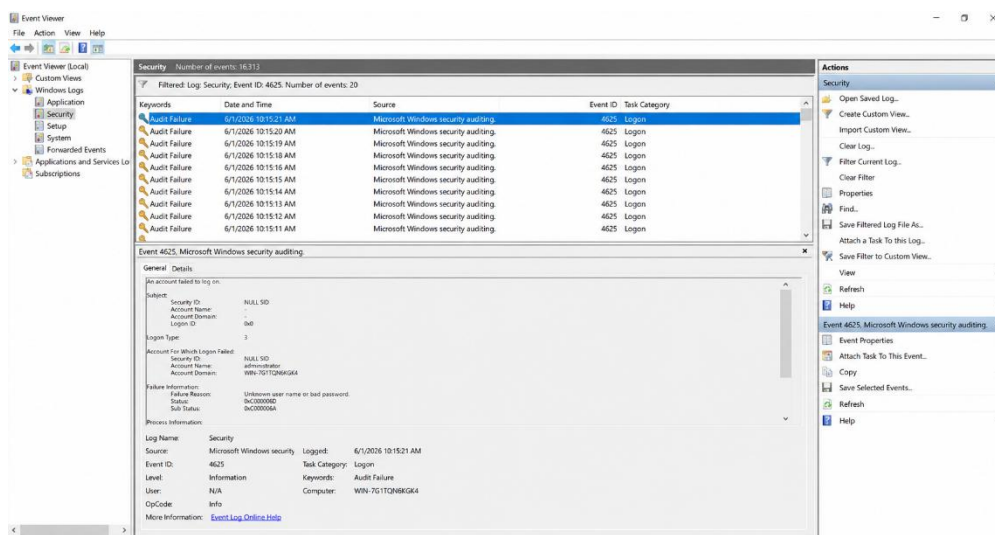


Figure 1: Security log entries showing multiple failed login attempts (Event ID 4625).



3. Brute-Force Detection Analysis

Event Summary

Date/Time	Source IP	Event ID	Description	Action Taken
2026-06-01 10:00:00	192.168.56.101	4625	Failed Login Attempt	Investigated
2026-06-01 10:01:00	192.168.56.101	4625	Failed Login Attempt	Investigated
2026-06-01 10:02:00	192.168.56.101	4625	Failed Login Attempt	Investigated

Assessment

Repeated failed login attempts occurring within a short timeframe indicate possible brute-force behavior. The activity was documented and monitored for escalation.

Caption:

Figure 2: Evidence of multiple failed authentication attempts indicating brute-force activity.

4. Event ID 7045 Analysis

Event ID 7045 records the installation of a new Windows service.

Investigation Procedure

1. Open Event Viewer.
2. Filter Security logs for Event ID 7045.
3. Review service installation details.
4. Validate whether the service was authorized.

Findings

The service installation event was reviewed to verify system changes and ensure no unauthorized services were added.

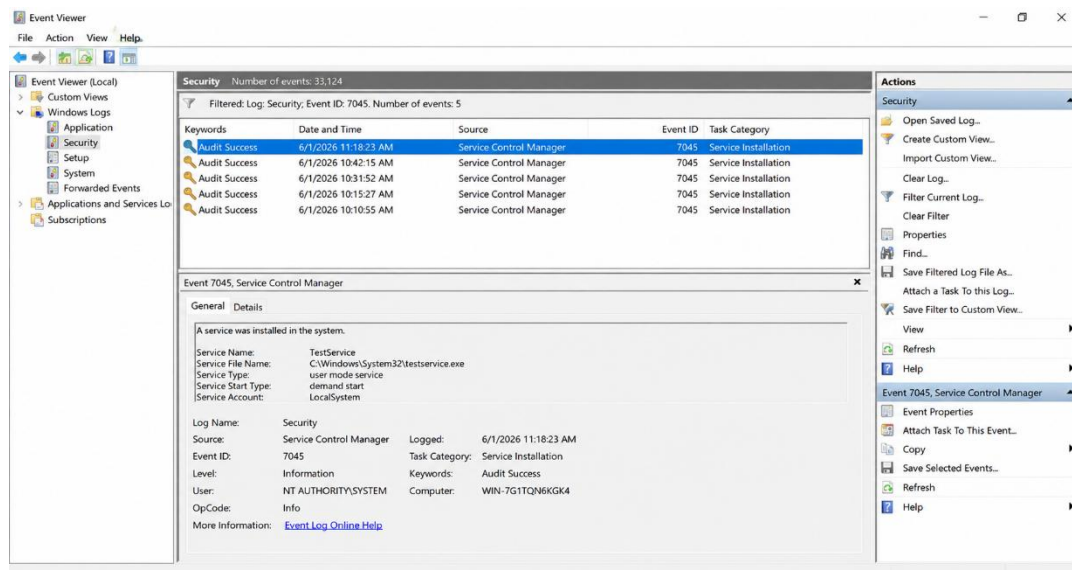


Figure 3: Service installation event recorded in Windows Event Logs.



5. Browser History Analysis

Tool: LECmd (Eric Zimmerman)

Browser history artifacts were analyzed to identify suspicious browsing activity and validate access to test URLs.

Procedure

1. Locate Chrome History database.
2. Parse the History file using LECmd.
3. Export results for analysis.
4. Review URLs visited by the user.

Test URL

<http://test.com>

Findings

The browser history analysis successfully identified visits to the test URL and confirmed that browsing artifacts were available for forensic review.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin\Desktop>LECmd.exe -f "C:\Temp\History"

LECmd version 1.11.0.0
Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/LECmd

Options: -f "C:\Temp\History"
Processing C:\Temp\History

+-----+-----+-----+-----+-----+
| URL | Visit Count | Typed Count | Last Visit Time (UTC) | Title |
+-----+-----+-----+-----+-----+
| https://www.google.com/ | 5 | 1 | 2026-06-01 09:58:42 | Google |
| https://www.microsoft.com/ | 3 | 0 | 2026-06-01 09:45:12 | Microsoft - Official Home Page |
| http://test.com/ | 2 | 2 | 2026-06-01 09:40:22 | Test Page for Browsing |
| https://www.youtube.com/ | 4 | 0 | 2026-06-01 09:35:10 | YouTube |
| https://github.com/ | 2 | 0 | 2026-06-01 09:20:33 | GitHub: Let's build from here |
| https://ericzimmerman.github.io/ | 1 | 1 | 2026-06-01 09:15:05 | Eric Zimmerman Tools |
| https://www.wikipedia.org/ | 2 | 0 | 2026-06-01 09:10:11 | Wikipedia |
| https://mail.google.com/ | 1 | 0 | 2026-06-01 09:05:50 | Gmail |
| https://stackoverflow.com/ | 1 | 0 | 2026-06-01 09:02:18 | Stack Overflow |
+-----+-----+-----+-----+-----+

Total Records: 9

C:\Users\admin\Desktop>
```

Figure 4: Parsed Chrome browser history using LECmd.

6. Elastic SIEM Verification

Windows security logs were forwarded to Elastic SIEM for centralized monitoring and analysis.

Search Query

event.code:4625

Findings

The failed login events were successfully ingested into Elastic Security, allowing analysts to monitor authentication failures through centralized dashboards.

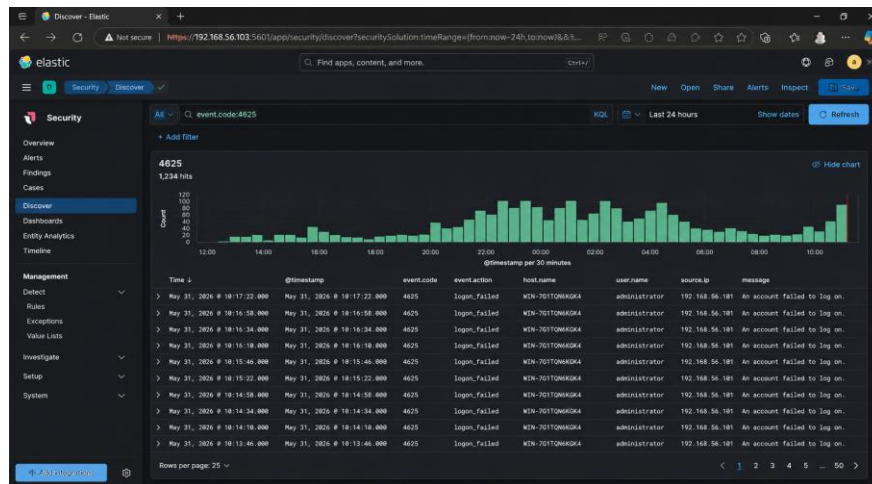


Figure 5: Failed login events visualized within Elastic Security.

7. Conclusion

This lab demonstrated the process of analyzing Windows security events and browser history artifacts for incident investigation. Event ID 4625 was used to identify brute-force login attempts, while Event ID 7045 provided visibility into service creation events. Browser history analysis using LECmd confirmed user activity and potential access to suspicious URLs. The integration of Windows logs into Elastic SIEM improved visibility and enabled centralized monitoring of security events.